

Paediatric Health — Child Health and Development

1. Child Development and Growth

Growth and development in childhood is a continuous process of physical, cognitive, language, social-emotional, and motor maturation. The first 1000 days of life (from conception to age 2) is the most critical window for neurodevelopment — interventions during this period have the greatest impact on lifelong health.

Growth monitoring: WHO Growth Standards (2006) are used globally. Plot weight, height/length, and head circumference on growth charts at every well-child visit. Key indicators: weight-for-age, length/height-for-age, weight-for-length, BMI-for-age. Z-scores: <-2 SD = undernutrition; <-3 SD = severe.

Development surveillance: use validated tools at each visit. Denver Developmental Screening Test II (DDST-II) or Ages and Stages Questionnaires (ASQ). Developmental milestones: social smile (6 weeks), head control (3-4 months), sitting independently (6-7 months), standing with support (8-9 months), walking (12-15 months), first words (12 months), 2-word phrases (18-24 months).

2. Immunisation Schedule in India

Age	Vaccines
Birth	BCG, OPV-0, HBV-0, Vitamin K
6 weeks	DTwP-1, IPV-1, HBV-1, Hib-1, PCV-1, RV-1
10 weeks	DTwP-2, IPV-2, HBV-2, Hib-2, PCV-2, RV-2
14 weeks	DTwP-3, IPV-3, HBV-3, Hib-3, PCV-3, RV-3
9 months	MMR-1, JE-1 (endemic areas), Typhoid (conjugate)
12 months	Hepatitis A-1, Varicella-1
15 months	MMR-2, Varicella-2
16-24 months	DTwP Booster, IPV Booster, Hib Booster
5 years	DTwP Booster-2
10-12 years	HPV (girls, 2 doses 6 months apart), Tdap

India's Universal Immunisation Programme (UIP) provides free vaccinations for all children. The programme has achieved near-100% coverage for OPV and BCG. PCV was introduced nationally in 2017. Rotavirus vaccine rollout completed in all states by 2019.

3. Common Childhood Illnesses

Acute Respiratory Infections (ARIs): leading cause of child mortality globally. Pneumonia: fast breathing, chest indrawing, grunting — requires antibiotics (amoxicillin). Bronchiolitis in infants: RSV most common cause — wheeze, hyperinflation, difficulty feeding. Mainly supportive treatment (oxygen, feeding support).

Acute Diarrhoeal Disease (ADD): second leading cause of child death under 5. ORS (Oral Rehydration Solution) is the cornerstone of treatment. Zinc (20 mg/day for 14 days) reduces severity and duration. Rotavirus vaccine dramatically reduces rotavirus diarrhoea hospitalisations.

Malnutrition and infection create a vicious cycle — infection worsens nutritional status; malnutrition increases infection severity. WASH (Water, Sanitation and Hygiene) is a fundamental preventive intervention.

Febrile convulsions: common (2-5% of children, 6 months-5 years). Usually brief (<15 min), generalised, with full recovery. No long-term risk in simple febrile seizures. Reassure parents. Antipyretics do NOT prevent recurrence. Complex febrile seizures (>15 min, focal, >1 in 24 hrs) need investigation.

4. Neonatal Care

Neonates (0-28 days) are particularly vulnerable. India's Neonatal Mortality Rate (NMR) is 20 per 1000 live births (NFHS-5). Leading causes: prematurity, birth asphyxia, neonatal infections (sepsis), neonatal jaundice, low birth weight.

Essential Newborn Care (ENC): immediate drying and warming, cord clamping at 1-3 minutes, initiation of breastfeeding within 1 hour, eye prophylaxis (tetracycline ointment), Vitamin K injection, BCG vaccine, birth dose HBV and OPV-0.

Kangaroo Mother Care (KMC): skin-to-skin contact between mother and preterm/LBW infant, exclusive breastfeeding, early discharge and close follow-up. WHO recommends KMC from

birth for preterm <2 kg. Reduces neonatal mortality by 40%.

Neonatal jaundice: physiological (common, peaks day 3-5, resolves by day 2 weeks).

Pathological: <24 hours (always pathological — haemolytic disease), prolonged (>2 weeks in term), very high bilirubin (kernicterus risk). Phototherapy for significant jaundice. Exchange transfusion for dangerous levels.

Newborn screening: Rashtriya Bal Swasthya Karyakram (RBSK) screens for congenital hypothyroidism (TSH heel-prick), sickle cell disease, and G6PD deficiency in selected states.

5. Nutrition in Children

Exclusive breastfeeding for 6 months, then complementary feeding with continued breastfeeding to 2 years, is the global recommendation. Breast milk provides all required nutrients, maternal antibodies, and growth factors. Colostrum (first days) is particularly rich in IgA.

Infant formula: for those who cannot be breastfed. Standard WHO/CODEX compliant formulas. Avoid cow's milk as a main drink under 12 months (low iron, high renal solute load, associated with iron deficiency anaemia).

Complementary feeding: introduce at exactly 6 months. Iron-rich first foods are critical (iron stores deplete at 6 months). Encourage dietary diversity (5+ food groups daily), avoid salt, sugar, honey (botulism risk <12 months), and fruit juice excess.

Vitamin A supplementation: India's National Vitamin A Supplementation Programme provides 100,000 IU Vitamin A at 9 months, then 200,000 IU every 6 months to age 5. Prevents vitamin A deficiency (VAD) — leading preventable cause of childhood blindness.

6. Diarrhoeal Disease and ORS

ORS (Oral Rehydration Solution): one of the greatest medical advances of the 20th century, credited with saving over 50 million lives. Low-osmolarity WHO ORS: sodium 75 mmol/L, glucose 75 mmol/L, potassium 20 mmol/L, chloride 65 mmol/L, citrate 10 mmol/L.

Homemade: 1 level tsp salt + 6 level tsp sugar in 1 litre clean water.

Zinc: 20 mg/day for 14 days in children >6 months, 10 mg/day for <6 months. Reduces diarrhoea duration by 25%, reduces subsequent episodes in following 2-3 months. Given

alongside ORS for all acute diarrhoea.

Antibiotics: NOT routine for acute watery diarrhoea. Indicated for: cholera (azithromycin), shigella (ciprofloxacin or azithromycin), confirmed Giardia or Entamoeba histolytica (metronidazole), or traveller's diarrhoea in high-risk.

WHO indicators: dehydration assessment. No dehydration: treat at home with extra ORS (50-100 ml after each loose stool). Some dehydration: ORS 75 ml/kg over 4 hours in health facility. Severe dehydration or IV fluid requirement (Ringer's Lactate or Normal Saline 30 ml/kg bolus over 30 min).

7. IMNCI — Integrated Management of Neonatal and Childhood Illness

IMNCI is the WHO/UNICEF evidence-based strategy for improving child survival. It provides a standardised approach to assess and classify illness in children aged 0-59 months, combining management of the major childhood killers: ARI, diarrhoea, malaria, malnutrition, and measles.

Assess-Classify-Treat algorithm: 3 zones (green — manage at home, yellow — refer to health facility, red — urgent referral to hospital with pre-referral treatment).

General danger signs in IMNCI: not able to drink or breastfeed, vomiting everything, convulsions, lethargic or unconscious — any one of these = urgent referral regardless of primary diagnosis.

F-IMNCI (Facility IMNCI): for first-level health facilities. Includes inpatient management, oxygen therapy, IV antibiotics, management of neonatal problems.

8. Child Mental Health and Developmental Disorders

Neurodevelopmental disorders: ADHD (Attention-Deficit/Hyperactivity Disorder), ASD (Autism Spectrum Disorder), intellectual disability, learning disabilities (dyslexia, dyscalculia). Early identification (typically in preschool/early school years) and multidisciplinary intervention significantly improve outcomes.

Red flags for autism: no social smile by 6 weeks, no babbling or pointing by 12 months, no two-word phrases by 24 months, any loss of previously acquired language or social skills at

any age. M-CHAT-R/F is a validated screening tool at 18-24 months.

Cerebral palsy: permanent, non-progressive motor disability from damage to the developing brain. Prevalence 2-3/1000 live births. Early physiotherapy, occupational therapy, speech therapy, and assistive technology are key interventions.

Child and adolescent mental health: depression, anxiety, eating disorders, and school refusal are increasingly recognised. India has a severe shortage of child psychiatrists (~150 for 1 billion-plus population). School-based mental health programmes, teacher training, and telepsychiatry are potential scale-up strategies.

9. Child Safety and Injury Prevention

Drowning is the leading cause of accidental death in children 1-14 years in India. Prevention: fencing around water bodies, swimming lessons from age 4, adult supervision near water, CPR training for parents and caregivers.

Road traffic injuries: a leading cause of child death and disability. Child restraint systems (car seats, booster seats) reduce injury by 70-80%. Properly fitted helmets for cycling. Pedestrian safety education. School zone speed limits.

Burns: children under 5 most vulnerable. Prevention: keep hot liquids away from edges, use back burners, supervise children in kitchen, lower water heater temperature to <48°C, smoke detectors in homes. Treat superficial burns with cool running water x 20 minutes.

Child abuse and neglect: physical, sexual, emotional abuse and neglect are globally prevalent. POCSO (Protection of Children from Sexual Offences Act) 2012 in India provides legal protection. Healthcare providers have a mandatory duty to report suspected child abuse. Child Helpline 1098.

10. Adolescent Health

Adolescence (10-19 years) is characterised by rapid physical, cognitive, and social-emotional development. Puberty — girls (thelarche 8-13 years, menarche 10-16 years); boys (testicular development 9-14 years).

Adolescent nutrition: iron and calcium requirements are highest during adolescence. Iron supplementation under WIFS programme. Calcium intake critical for bone density — dairy,

ragi, sesame seeds. Avoid crash diets and eating disorders.

Reproductive and sexual health: comprehensive sexuality education reduces STI risk and unwanted pregnancies. Menstrual hygiene management — availability of sanitary products, changing facilities, and destigmatisation. India's Kishori Shakti Yojana and Rajiv Gandhi Scheme for Empowerment of Adolescent Girls (SABLA) address adolescent girl health.

RKSK — Rashtriya Kishor Swasthya Karyakram: India's national adolescent health programme. Provides services through Adolescent Friendly Health Clinics (AFHC) at PHC/CHC level for nutrition, sexual and reproductive health, substance abuse, NCDs, mental health, violence, injuries.

India has the world's largest adolescent population (243 million, 10-19 age group). Investing in adolescent health yields enormous dividends — better educational outcomes, lower teenage pregnancy, reduced burden of NCDs in adult life, and greater economic productivity.